

# Driving in the wrong direction? A co-evolutionary model of electric vehicle adoption and innovation

## Abstract

Car-centric societies face substantial challenges in moving towards sustainable mobility systems, with internal combustion engine vehicles remaining a major source of emissions. Electric vehicles play a critical role in addressing this challenge, yet their diffusion depends on the interaction of consumer behaviour, firm innovation, and policy incentives. This paper develops an agent-based model to examine these dynamics, calibrated on the data for the state of California over 2001-2023. In the model, heterogeneous car users influenced by their social peers purchase cars, both new and used, based on discrete choice theory, while manufacturers develop new car models, adjusting prices accordingly. We examine how different policy combinations can achieve deployment targets while balancing economic costs, emissions and consumer utility. Our simulations compare individual and combined policy instruments, including carbon pricing, new and used car purchase rebates, production subsidies and electricity price subsidies. Results indicate that only carbon pricing and new vehicle purchase rebates achieve a target of 95% electric vehicle adoption by 2035. Each policy has key trade-offs: carbon pricing substantially reduces consumer welfare, while new purchase rebates create heavy fiscal burdens. Policy combinations prove to be more effective by reducing the intensity of the required interventions and minimising the induced rebound effect and trade-offs. Extended simulations till 2050 demonstrate that EV adoption rates stabilize after policy withdrawal. However, cumulative emissions in the business-as-usual scenario are lower than those of a policy-induced mobility transition until the 2040s, indicating that additional policy is required to reduce reliance on private vehicles.

Keywords: policy-mix design, threshold model, carbon price, electricity subsidy, social network

## 1 Introduction

In 2019, 10% of global greenhouse gas emissions were attributed to road transport, which represented 70% of all carbon emissions related to transport. Therefore, the environmental sustainability of future generations requires large scale reductions in the carbon footprint of road transport. The large-scale adoption of electric vehicles (EVs), to replace internal combustion engine vehicles (ICEVs), is an indispensable condition for the transition to sustainable mobility in car-dependent urbanizations (Jaramillo et al., 2022, p 1078). This requires policy incentives to be cost-effective, environmentally sustainable and accepted by the public. However, the co-evolving socioeconomic interactions between consumer adoption of new products and producer investments in innovations make difficult for policy makers to anticipate the outcomes of the policies before implementing them. This is relevant not only for designing policies that are effective in promoting

electric vehicle adoption, but also to avoid unintended negative consequences on other variables (Herrmann and Savin, 2017; Nuñez-Jimenez et al., 2022).

Agent-based simulation models (ABMs) are an appealing method to address public policy issues due to their bottom-up structure that accounts for agent heterogeneity and the complex interplay of individual behaviour. These models have offered useful policy design laboratories for stimulating the diffusion of EVs (Domarchi and Cherchi, 2023; Mehdizadeh et al., 2022), often calibrated for the data of the United States (Brown, 2013; Noori and Tatari, 2016; Adepetu et al., 2016; Sun et al., 2019; Spangher et al., 2019; Ledna et al., 2022). In the context of mobility transitions, the study of social networks and opinion formation dynamics has increased in popularity (McCoy and Lyons, 2014; Silvia and Krause, 2016; Li et al., 2019; Feng et al., 2019; He et al., 2020; Ning et al., 2020; Lee and Brown, 2021; Zhang et al., 2022). The literature of ABMs for EV diffusion also provides studies addressing the role of directed innovation in promoting more sustainable modes of transportation (Windrum et al., 2009; Greene et al., 2014; Sun et al., 2019; Fan and Li, 2025).

Despite the growing literature, the interaction between social dynamics and innovation decisions remains underexplored. Specifically, the interplay between consumer adoption and firm innovation presents important considerations for policy analysis due to their co-evolving nature. Firms direct their innovation efforts to electric vehicles based on the size of its customer base. In turn, people are willing to adopt EVs if their attributes and price can compete with ICEVs. Without policy intervention, this interaction slows down the deployment of new products. However, policies motivating the early adoption and investment of EVs can trigger a virtuous circle that accelerates the transition. In order to be sustainable, such transition requires minimising policy costs, welfare losses, and greenhouse gas rebound effects. Therefore, our paper addresses the following research question: What policy combinations that meet adoption targets can best balance the trade-offs between costs, consumer utility, and emissions?

This paper provides a framework to support policy design accounting for the co-evolution of adoption and innovation. We introduce an ABM combining a multinomial logit model for consumer choices (Østli et al., 2017), a social imitation model for the diffusion of electric vehicles (Rogers, 2003), and an  $NK$  model for the directed innovation efforts of manufacturers (Windrum et al., 2009). These elements allow capturing the interaction between social imitation among heterogeneous car users and profit-oriented investment choices. Furthermore, we include a stylized representation of the market for used cars given its paramount importance in the US car market.

To calibrate the model, we set parameters to replicate the conditions of the state of California from 2001 to 2023, due to its rich data availability on mobility. Furthermore, California is an example of a car-dependent territory committed to a reduction in carbon emissions as part of its Zero-Emission Vehicle (ZEV) requirement, adopted by the California Air Resources Board. This mandate is complemented by Federal and State rebates supporting the adoption of EVs.

The model is used to test policy scenarios for the period 2024-2035. Particularly, we test a carbon price, a rebate for new and used EVs, an electricity subsidy and an EV production subsidy. Policies are evaluated in their ability to reach a target adoption rate and compared on their impact on total net policy costs, cumulative carbon emissions, and consumer utility. In this way, we study the induced trade-offs associated with each policy instrument. Moreover by combining the instruments in pairs we study how their performance can be improved. Finally, we show the stability of the EV transition by

simulating for the period 2036-2050 after removing policy intervention.

The remainder of the paper is organized as follows. In Section 2 we formulate the automotive sector model of endogenous innovation and social imitation in the context of climate policy. Section ?? provides the details of our calibration strategy to externally validate the model outputs with respect to Californian data. In Section ??, we analyze the results of the model. Finally, Section ?? concludes and outlines future research avenues.

## 2 The model

### 2.1 Overview

The present model studies policies for the diffusion of EVs in the context where innovation and social imitation are co-evolving processes. The model features a market for new and used cars that are characterized by a car type (ICEV or EV), production cost, fuel efficiency, battery or fuel tank size, and an abstract quality measure. The model is divided into four submodules: the discrete choice consumption submodule, the social imitation submodule, the innovation submodule and the used car market submodule. Figure ?? provides an overview of the interactions between each submodule. The model is in discrete time, where each time step represents a month. For model equations, see Appendix ??.

For the representation of the users' choice of car, we use a discrete choice logit model (McFadden, 1974), as in previous studies addressing the adoption of EVs (Eggers and Eggers, 2011; Shafiei et al., 2012; Østli et al., 2017). Heterogeneous car users meet their mobility needs either keeping their current car, purchasing a new or a used car. The evaluation of a car is based on its attributes and consumer preferences. We add to this literature by modelling car users that are forward-looking in estimating the lifecycle costs and emissions when evaluating a car.

We use the diffusion of innovations theory by Rogers (Rogers, 2003) to conceptualize the willingness to consider EVs. Users become willing to consider buying an EV after a sufficient number of related peers are using it. The number of neighbours threshold varies amongst car users, with its value corresponding to a spectrum between "innovators" and "laggards".

Car manufacturers select the car designs to put on sale based on their profit expectations throughout the lifecycle of the technology, evaluated at the prices consistent with profit maximization. New ICEV or EV vintages are discovered as a result of innovation. Following similar studies (Windrum et al., 2009), we employ the *NK* model (Kauffman, 1993) to represent the distribution of possible technologies. In this setting, the available car designs depend on previous *R&D* choices, leading to specialization patterns and emergent heterogeneity.

For the representation of the market for used cars, we propose a stylized model where used car dealers are consolidated into a single entity. The price of used cars follows the price of their respective most similar new car and applies a price discount proportional to the car age. Whenever a consumer buys a car, the replaced car is sold to the used car dealer at its market price with a discount. Used car dealers discard cars whose market price falls below its scraping value.

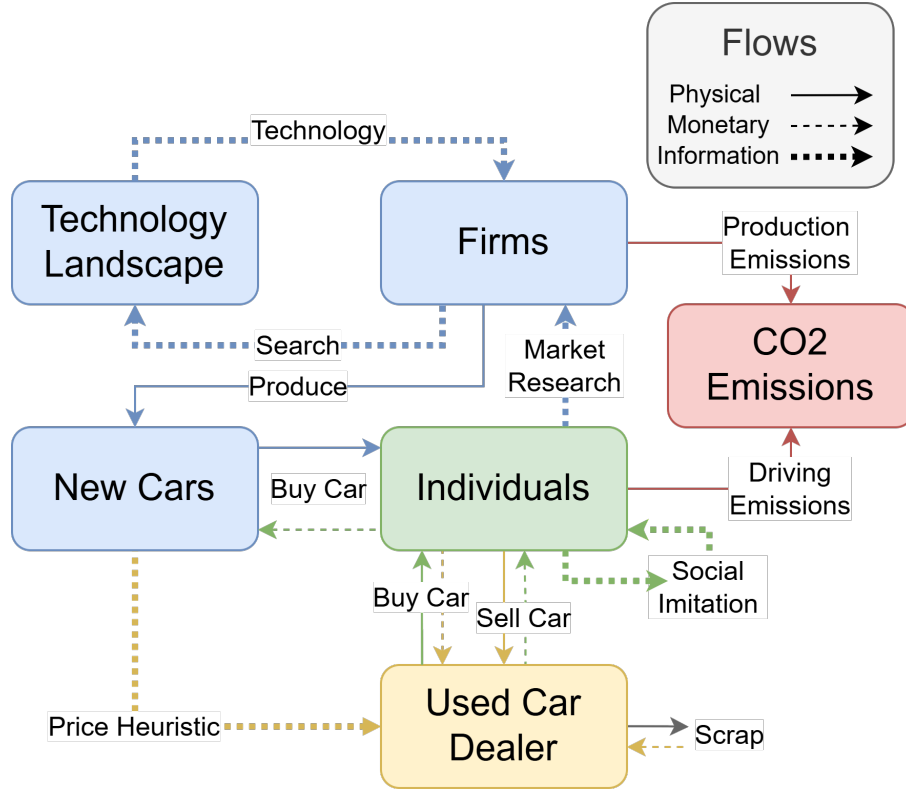


Figure 1: Model structure

## 2.2 Car users

Car users drive a fixed and heterogeneous distance each month. Stochastically, users are sequentially activated to replace their car for one of new or used cars available on the market. Following a social imitation model where car users are placed in a small-world network, the willingness to consider an EV is updated at the end of the period. The model assumes one car for each driver, as the US car-to-driver ratio has remained stable at 1.15 from 2012 to 2023 ([Bureau of Transportation Statistics](#)).

### 2.2.1 Car choice

Car users compare the lifecycle utility of their own car with that of the available used and new cars the user is willing to consider. The utility of a car is given by the willingness to pay for its quality and range, each of them with diminishing returns, minus its lifecycle costs and emissions (Jaramillo et al., 2022, p.4). The lifecycle costs are given by the sum of the purchase price, net of the sale of the user's current car, and the discounted sum of the expected cost per kilometre. The lifecycle emissions are given by the manufacturing emissions if the car is new and the discounted sum of the expected emissions per kilometre. When estimating future costs and emissions related to driving, car users factor in the depreciation of fuel efficiency taking place at the end of each time-step at a fixed rate. Given the uncertainty in gasoline and electricity markets, car users assume the future prices and emissions of gasoline and electricity to be the same as in the present period.

Given a utility for each of the alternative car choices (including their own) car users select one using a logit model (McFadden, 1974) where the probability of selecting a car is proportional to its assigned utility.

### 2.2.2 Social imitation

To represent the influence of peers on the choice of car type, we build on the threshold models of EV adoption outlined in Eppstein et al., 2011; McCoy and Lyons, 2014; Wang et al., 2023. We consider a social network in which car users change their car purchasing behaviour based on the proportion of EV owned by their neighbours. This threshold is heterogeneous amongst car users, with a low threshold representing "innovators" who are willing to adopt new technologies, whilst a high threshold corresponds to "laggards" who will only consider moving away from the established technology once it is very common in their network neighbourhood.

The result of the threshold model is that car users may not adopt EVs even if those have high utility. This structure captures the gap between behaviour and attitudes or beliefs in which car users may have sufficient desire (high utility) to pursue pro-environmental behaviours but do not do so due to other limiting factors (Kollmuss and Agyeman, 2002), such as the lack of an established social norm. Thus, it is possible for car users to have a high willingness to pay for lower emissions and high emissions due to driving an ICEV. On the other hand, with policy intervention, social imitation dynamics can amplify the policy effect by increasing the visibility of EVs (Konc et al., 2021).

Social interaction is structured using a small world network (Watts and Strogatz, 1998) to replicate word-of-mouth social interactions in highly clustered social circles where neighbours are also likely to become friends. This network is formed by first arranging car users in a ring, with each person connected to a homogenous number of nearest neighbours, a fixed number of connections each to their nearest neighbours. Then the symmetry of the network is broken up by introducing long-range weak ties across the network through probabilistic re-wiring of connections.

## 2.3 Car supply

Manufacturers sell multiple car designs that are produced on demand. Stochastically, they update the car types on sale and their respective prices in line with profit maximization. For simplicity, we assume car manufacturers face constant production costs and no capacity or financial constraints. With these assumptions, we ignore some relevant aspects for the transition, such as carbon lock-in (Unruh, 2000; Unruh, 2002) or lack of financial appetite from venture capitalists (Mazzucato, 2013; Mazzucato and Semieniuk, 2017). We choose this strong assumption based on the fact that the car industry is formed by firms that are integrated in global supply chains and financial markets, while our model is restricted to the local market.

Following the Segmentation, Targeting and Positioning framework (Kotler, 1989), manufacturers supply car designs targeting a specific consumer segment. A consumer segment is defined as a relatively homogeneous group of customers with particular needs and preferences (Lynn, 2012). In this model, segments are formed on the basis of EV acceptance, willingness to pay for quality and willingness to pay for emissions reduction. These are relevant determinants of car choice after controlling for socio-demographic attributes (Axsen and Kurani, 2013; Keyes and Crawford-Brown, 2018; Jang and Choi, 2021).

### 2.3.1 Technology choice and pricing

Manufacturers evaluate the expected profit at each consumer segment of each car type using known technologies. The expected profit is given by the marginal profit, the segment population, and the expected market share within the segment. This expected in-segment market share depends on the attributes of the car and the state of competition in that segment (over the last 12 months). Crucially, it is also consistent with the discrete choice consumption model, where the price is defined to maximize expected profits.

The choice of car types being supplied is resolved heuristically. Manufacturers include in their supply line the most profitable car targeted at a specific segment. Once the segment is satisfied, the selected price is locked. The remaining segments are satisfied by sequentially selecting the most profitable car, where the expected profits of unselected cars targeted at their most profitable remaining segment are compared to that of the selected cars with the profit-maximizing price for the selected segments. As a result, a car may be produced to satisfy multiple segments at a single price if it outperforms other designs.

By assuming that all manufacturers target all segments, we abstract from the brand-reputation considerations. This simplification allows us to focus on the interaction between innovation and adoption, especially in the context of transition policies in the car industry. However, to encourage specialization into specific segments, we limit the number of cars that may be produced by a single manufacturer to below the total number of segments and limit the number of technologies a manufacturer can retain in its memory.

### 2.3.2 Innovation

Stochastically, car manufacturers innovate and learn technology to produce a new car type with unique characteristics. We use  $NK$  models to conceptualize innovation (Levinthal, 1997; Rivkin, 2000; Baumann and Siggelkow, 2013). In this framework, technologies are distributed on a map where local movements lead to incremental improvements in product attributes.  $NK$  models allow the representation of complex technological landscapes with multiple local solutions, accounting for the path-dependent nature of innovation. This assumption allows the capture of heterogeneous abilities to adopt technologies, or absorptive capacity (Cohen and Levinthal, 1990), which are gained with path-dependent experience (Nelson and Winter, 1977).

Manufacturers are engaged in a stochastic parallel search in the landscapes for ICEVs and EVs. We limit innovation to the development of one single car, thus making the choice to develop ICEVs or EVs exclusive. This choice is made based on the profit expectation of using the candidate car, whose attributes are known to the manufacturer, priced at its most profitable segment. Mirroring consumer car choice, firms use a logit model to choose which innovative design to pursue, with an intensity of choice parameter  $\lambda$  that determines the balance of exploration and exploitation (Puranam et al., 2015).

As in previous studies (Jain and Kogut, 2014), manufacturers discard previously searched solutions when forming the set of candidate technologies to explore. This set contains all the unexplored neighbouring technologies with respect to the past search location on each landscape. We include the past search location, allowing the possibility of not searching if there are no attractive alternatives. Once a new design has been researched and the memory capacity is exceeded, the oldest car design not in production is forgotten. The limited memory capacity is a cognitive limitation whereby the searching agent economizes cognitive power by forgetting unused solutions.



### 2.3.3 Used cars

The market for used cars represents the bulk of car purchases in the US<sup>1</sup>, providing an affordable alternative to low-income users. This model chooses a parsimonious approach for the representation of the market for used cars. Used car dealers are consolidated into a single entity that sets the purchase and sale prices of used cars.

The sale price of a used car is a fraction of the price of its most similar car on the market for new cars. The fraction is inversely proportional to the used car age, where the price depreciation rate of EVs is larger than that of ICEVs (Schloter, 2022). The used car purchase price is a fraction of its sale price, given by a fixed target mark-up that serves as a proxy for the degree of monopsony in the market for used cars. Sale and purchase prices have an exogenous lower bound representing the car scrap value. Cars with prices lower than their scrap value are removed from circulation. To reduce computational cost, we impose a maximum capacity on the second-hand market after which the cars with the lowest value are removed.

## 3 Calibration and validation

We calibrate the model to the data for the state of California 2001 to 2023, where each model time step represents a month. For more details on data sources and justification, see Supplementary Material. All monetary units are expressed in 2020 US dollars. The price and emissions per kilowatt hour of gasoline and electricity during the validation period reflect real world monthly data, while the distribution of monthly distance driven is taken from survey data. In the case of distance travelled, we assume no rebound due to a change in car type (Zhang et al., 2025). We extrapolate secondary data from previous studies to calibrate the willingness to pay for range increases and emissions reductions, the boundaries of the car attributes in the NK landscape (except for quality), the car scrap value, the discount rate, and the price depreciation rate of used cars.

For other non-observable parameters, we follow past history-friendly models (Malerba et al., 2008; Herrmann and Savin, 2017) in avoiding excessive calibration. Instead, we fix the least sensitive parameters to some value at a reasonable order of magnitude and perform the respective sensitivity analyses afterwards. This applies to the parameter governing technological complexity, the research intensity of choice, the mark-up applied in the used car market, and the diminishing returns to utility with respect to quality. To capture the nuances between EV and ICE car production, we assume the battery size is correlated to unit production costs by a coefficient of .5, such that increases in EV range may come at a greater cost.

The remaining parameters are indirectly calibrated to replicate the trend of EV uptake and percentage of EV sales in California (see [California Energy Commission](#) data). Additionally, our indirect calibration keeps output variables such as prices, market concentration or average car age within the observed ranges described in Table ???. In Figure ??, we show a representative distribution of the willingness-to-pay parameters, distances and innovativeness values.

Based on the conducted Sobol sensitivity analysis (see the Supplementary Material), the key parameters affecting our target variables are the distributions of the willingness to pay for quality and of the imitation threshold, the consumer intensity of choice, and the

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Table 1: Target variable ranges

| Variable                  | Range                                   | Source                              |
|---------------------------|-----------------------------------------|-------------------------------------|
| Price for new cars        | 25th %: \$32,359.41, 75th %:\$57,784.66 | (Grieco et al., 2024, Figure II)    |
| Herfindahl-Hirshman index | (.11-.18)                               | (Grieco et al., 2024, Figure II)    |
| Average car age between   | (10-12) years                           | Bureau of Transportation Statistics |

fuel efficiency depreciation rate. Since the willingness to pay for the quality parameter mostly affects the price level, we determine its median value analytically in such a way that the optimal price of an ICEV with average features targeted at a consumer with average willingness to pay equals the observed average price. Then, we distribute this value according to the Californian income distribution, assigning a higher willingness to pay to richer agents.

For the users’ intensity of choice and the depreciation rate of car efficiency, we perform a grid search to match the stylized facts of car price range and age, iteratively narrowing the search space until matching the trends for EV uptake while keeping the other target variables within their observed ranges. Lastly, a beta distribution is used to define the heterogeneous values of car users’ EV consideration thresholds. Given the high computation cost of the ABM, simulation-based Bayesian inference (Papamakarios and Murray, 2016; Cranmer et al., 2020; Dyer et al., 2024) is used to calibrate the beta distribution values to match EV uptake during the validation period.

We simulate a population of 3,000 car users and 16 car manufacturers (Grieco et al., 2024). The model simulates a burn-in period of 15 years (180 time-steps) before the validation period. The model is initialized with car manufacturers being placed at one Hamming distance from the worst technology when evaluated at the median segment, and car users owning a random car among those manufacturers can produce. During the burn-in period, manufacturers improve and sell ICEVs, while EV search and production is not permitted until the validation period.

In order to account for existing EV support policies, we feature a simplified version of the federal and state rebate assigned to EVs (California Air Resources Board, 1990a; California Air Resources Board, 1990c; California Air Resources Board, 1990b) from 2010-23, discounting a flat amount of \$10,000 to new EVs and \$1,000 to used EVs. In addition, the implemented carbon taxes are already included in the imported prices for gasoline and electricity. However, for the sake of simplicity, we do not model the existing fuel efficiency standards or the EV sale mandates stipulated in the ZEV Californian plan.

Figure ?? shows the model projections for the validation on EV uptake and percentage of EV sales (top left), prices of new and used EVs and ICEVs (top right), market concentration (bottom right), and average car age (bottom left). The simulated values are the average over 64 Monte Carlo simulations. These plots validate our indirect calibration exercise, given that the projections of EV uptake and sales closely match the real data and that prices, market concentration, and average car age are mostly within the ranges described in Table ??.



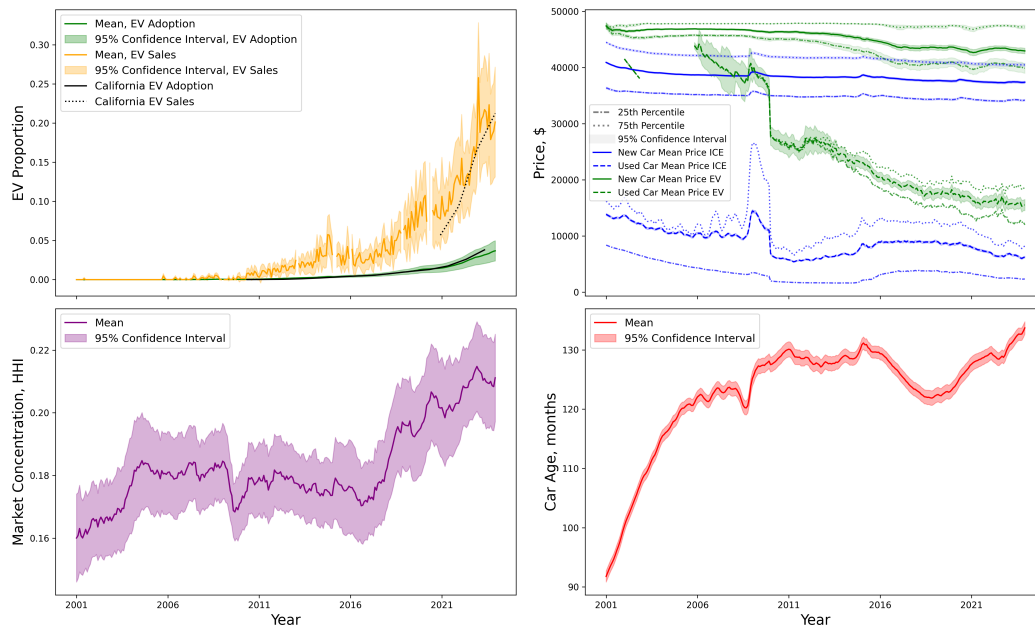


Figure 2: Simulation of key model outputs from 2001 to 2023. EV uptake and EV sales, in percent (top left); Retail prices for new and used ICEVs and EVs, in dollars (top right); Herfindahl-Hirschman Index, in percent (bottom left); and, average car age, in months (bottom right).

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## A Model Equations

### A.1 Car users

The set  $I$  collects all car users, each of them being indexed by  $i$ .

#### A.1.1 Car choice

With probability  $\eta \in (0, 1)$ , car users are activated to consider replacing their car. Given that cars in the second hand market can only be bought once, consumers are activated sequentially in a random order to make car choices and the stock of used cars is updated accordingly.

The alternatives a user considers include the users’ own car, all new and used ICEVs on sale, and all new and used EVs if the user considers them. Equation ?? defines the

set of cars a user considers as alternatives ( $A_{i,t}$ ):

$$A_{i,t} = \begin{cases} \{v \in V_t : v_{v,t} = i \vee v_{v,t} = 0\} \cup Y_t & \text{if } x_{i,t-1} = 1 \\ \{v \in V_t : (v_{v,t} = i \vee v_{v,t} = 0) \wedge z_{v,t} = 0\} \cup \{y \in Y_t : z_{y,t} = 0\} & \text{if } x_{i,t-1} = 0 \end{cases} \quad (1)$$

Where the set  $V_t$  defines the stock of active cars and the pointer  $v_{v,t}$  returns the car users' ID, or 0 if the car is in the market for used cars. The set  $Y_t$  contains all new car designs on sale. A crucial difference is that, given that this model assumes production takes place on demand, each new car designs on sale is represented by a vector of attributes and a sale price, but does not represent an actual car. Only when a new car is bought the car of a given design  $y \in Y_t$  is produced and included onto the set of active cars  $V_t$  with a unique ID  $v$ . The binary statements  $z_{v,t}$  or  $z_{y,t}$  respectively indicate that the car  $v$  or the design  $y$  corresponds to an EV, while the binary statement  $x_{i,t}$  indicates that the user  $i$  considers EVs. Note that, with probability  $1 - \eta$ , the set of considered alternatives only includes the users' own car.

Equation ?? describes the utility a user derives from a car alternative:

$$U_{a,i,t} = \beta_i Q_{a,t}^\alpha + \nu(B_{a,t} \omega_{a,t} (1 - \delta_{a,t})^{L_{a,t}})^\zeta - \gamma_i LCE_{a,i,t} - LCC_{a,i,t} \quad (2)$$

$$= \begin{cases} -P_{a,t} + \hat{P}_{i,t} - \gamma_i E_{a,t} + \beta_i Q_{a,t}^\alpha + \nu(B_{a,t} \omega_{a,t} (1 - \delta_{a,t})^{L_{a,t}})^\zeta - D_i \frac{(1+r)(1-\delta_{a,t})(c_{a,t} + \gamma_i e_{a,t})}{\omega_{a,t}(r - \delta_{a,t} - r\delta_{a,t})} & \text{if } v_{a,t} \neq i \\ \beta_i Q_{a,t}^\alpha + \nu(B_{a,t} \omega_{a,t} (1 - \delta_{a,t})^{L_{a,t}})^\zeta - D_i \frac{(1+r)(1-\delta_{a,t})(c_{a,t} + \gamma_i e_{a,t})}{\omega_{a,t}(r - \delta_{a,t} - r\delta_{a,t})} & \text{if } v_{a,t} = i \end{cases} \quad (3)$$

The parameter  $\beta_i$  represents the idiosyncratic willingness to pay, in dollars, a user assigns to the car quality ( $Q_{a,t}$ ) with diminishing returns scaled by  $\alpha > 0$ . Similarly, the parameter  $\nu$  denotes the dollar willingness to pay for car range with diminishing returns scaled by  $\zeta > 0$ . The range of a car is expressed as the product of its fuel tank (ICEV) or battery size (EV), denoted by  $B_{a,t}$  and expressed in KWh, and its fuel efficiency, denoted by  $\omega_{a,t}$  and expressed in kilometres per KWh. We assume a car's fuel efficiency suffers from depreciation at a fixed monthly rate  $\delta_{a,t}$ , which is specific to the car type (ICEV or EV). In calculating the present fuel efficiency of a car, we express the car age in months with the variable  $L_{a,t}$ .

The parameter  $\gamma_i$  represents the idiosyncratic willingness to pay, in dollars per kilogram of CO<sub>2</sub>, for carbon emissions reductions. In the utility derivation, this multiplies the expected lifecycle emissions of a car, expressed by  $LCE_{a,t}$ . These can be decomposed by the type-specific manufacturing emissions ( $E_{a,t}$ ), which are only considered if the evaluated car is new, and the expected lifecycle emissions from fuel or gasoline use. These depend on the user's idiosyncratic monthly distance driven, expressed in kilometres per month and denoted by  $D_i$ , on the car's fuel efficiency, and on the emissions per kilowatt hour of gasoline (ICEV) or electricity (EV), denoted by  $e_{a,t}$ , whose expected values into the future equal its most recent value. Emissions resulting from future driving behaviour are discounted in the utility derivation by a monthly discount rate  $r > 0$  (see the Supplementary Material for derivation).

Similarly, consumers consider the lifecycle costs of a car, which are denoted by  $LCC_{a,t}$ . The variable  $P_{a,t}$  denotes the purchase price of the evaluated car, taking a value of zero if the car is owned by the user. If the user is evaluating a car other than its own, it deducts from the purchase the price at which it can sell its own car, denoted by  $\hat{P}_{i,t}$ . The user also calculates the discounted cost from gasoline or electricity use, which has the same form as the discounted driving emissions and uses the naive expectations of the cost per kilowatt hour of gasoline or energy, expressed by  $c_{a,t}$  (see the Supplementary Material for derivation).

The pointer  $\psi_{i,t}$  returns the car chosen by a user from its considered set of alternatives. The probability an alternative is chosen is proportional to its associated derived utility relative to



the other alternatives, as described in Equation ??:

$$Pr(\psi_{i,t} = a' \in A_{i,t}) = \frac{\exp(\kappa U_{a',i,t})}{\sum_{a=0}^{A_{i,t}} \exp(\kappa U_{a,i,t})} \quad \text{if } v_{a',t} \neq i \quad (4)$$

Where  $\kappa > 0$  indicates the intensity of choice. A value of  $\kappa = \inf$  indicates users are utility maximising, whilst on the other extreme, for  $\kappa = 0$ , they choose cars randomly with equal probability.

### A.1.2 Social imitation

After all car ownership choices have been made, users re-evaluate their willingness to consider an EV. This is governed by the idiosyncratic imitation threshold  $\chi_i \in (0, 1)$ , expressed as a minimum share of adopter neighbours required for the user to consider adopting. The population subset  $H_i \subset I$  collects user  $i$ 's neighbours. This set is obtained from a Watts-Strogatz network with the small-world property to replicate word-of-mouth social interactions in highly clustered social circles where individual neighbours are also likely to be connected. We denote with  $V_{i,t}^H \subset V_t$  the subset of active cars that belong to car users in  $i$ 's network, as defined in Equation ??.

$$V_{i,t}^H = \{v \in V_t : v_{v,t} \in H_i\} \quad (5)$$

Equation ?? uses this set to determine whether an individual considers buying EVs, as a function of the share of EV users in  $i$ 's network:

$$x_{i,t} = \begin{cases} 1 & \text{if } \frac{|V_{i,t}^H : z_{v,t}=1|}{|H_i|} \geq \chi_i \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

## A.2 Car supply

The set  $J$  collects all firms, each of them being indexed by  $j$ .

### A.2.1 Market segmentation

Amid consumer heterogeneity in a multi-attribute market, firms segment the market into groups of representative consumers to effectively choose the features of their products and the optimal prices. They do so by building a representative consumer profile for each segment. Each segment  $s \in S$  is characterized by preference parameters  $(\beta_s, \gamma_s, \nu_s)$  and the binary statement indicating that the segment considers buying EVs ( $x_s$ ). The preference parameters correspond to the R-tile value of the population, where the parameters  $R^\beta, R^\gamma$  and  $R^\nu$  indicate the number of partitions in the parameter space. The population subset  $I_{s,t} \subset I$  includes the users whose preferences correspond to those defined in segment  $s$ .

### A.2.2 Technology choice and pricing

The set of technologies to produce car designs known to a manufacturer is denoted by  $M_{j,t}$ , which is a subset of all existing technology  $M$  defined by the NK landscapes for ICEVs and EVs ( $M = M^{ICE} \cup M^{EV}$ ) (see Supplementary Material). Each technology  $m$  is represented by a vector of car attributes and its corresponding coordinates in the NK landscape.

The estimated profit in segment  $s$  using a technology  $m$  at some price  $P$  is denoted by  $\Pi_{m,s,t}$  and defined in Equation ??:

$$\Pi(P)_{m,s,t} = \frac{(P - C_{m,t})|I_{s,t}|\exp(\kappa U_{m,s,t})}{\exp(\kappa U_{m,s,t}) + W_{s,t}} \quad (7)$$

Where  $U_{m,s,t}$  is the lifetime utility the car representative consumer with preferences  $\beta_s, \gamma_s$ , the unit production costs are denoted by  $C_{m,t} > 0$  and  $W_{s,t}$  indicates the expected state of competition in sector  $s$ . This indicator equals the moving average over the last 12 months of the sum of lifetime utilities in the new car market adjusted by the intensity of choice, as described in Equation ??:

$$W_{s,t} = \begin{cases} \sum_{t-13}^{12} \sum_{m \in Y_t: z_{y,t}=0} \exp(\kappa U_{m,s,t})/12 & \text{if } x_s = 0 \\ \sum_{t-13}^{12} \sum_{m \in Y_t} \exp(\kappa U_{m,s,t})/12 & \text{if } x_s = 1 \end{cases} \quad (8)$$

Notice that for segments that serve clients who do not consider EVs, only ICE cars are included in this metric.

For each segment and each known technology, firms define an optimal price ( $P_{m,s,t}^*$ ) that satisfies expected profit maximization, as defined in Equation ?? (see the Supplementary Material for derivation):

$$P_{s,m,t}^* = C_{m,t} + \frac{1}{\kappa} + \frac{\mathbb{W}\left(\exp(\kappa U_{m,s,t}(P = C_{m,t})) - 1\right) / W_{s,t}}{\kappa} \quad (9)$$

Thus, the optimal profit function may be broken down into 3 components: the production cost, a bounded rationality aspect which tends to 0 as car users become perfectly rational ( $\kappa \rightarrow \infty$ ) and a market advantage component which increases with the superior of a car technology over the state of the past market competition in that segment.

Firms update their supply line and prices with probability  $\theta \in (0, 1)$ , accounting for frictions in production, price rigidities, or delays in obtaining information on the segment population. When making supply decisions, firms use at most one technology per segment, a simplification justified by strategic considerations to avoid intra-firm competition. Choosing the product mix and the target segments that determine prices is a complex problem that firms resolve heuristically. Firms rank potential cars within each segment using a car-segment-specific optimal price. They then iteratively select the most profitable car-segment combination, fixing its price across all relevant segments before moving to the next best option. As a result, a car may be produced for multiple segments at a single price if it outperforms other choices. Formally, define a  $S \times |M_{j,t}|$  matrix containing the expected segment profit for each known technology ( $M_{j,t} \subset M$ ). Car variants are sequentially added into  $Y_{j,t}$  and accordingly priced using the algorithm below:

1. Activate algorithm with probability  $\theta$ , else end.
2. Denote with  $m'$  and  $s'$  the column  $m$  and the row  $s$  corresponding to the cell with the highest expected profit.
3. Include an element  $y$  in  $Y_{j,t}$  with the features of technology  $m \in M_{j,t}$  and the price  $P_{s',m',t}^*$ .
4. Delete the row  $s'$ .
5. For the elements in column  $m'$ , lock the price at  $P_{s',m',t}^*$  and update the expected profit values accordingly ( $\Pi_{s,m',t}(P_{s',m',t})$ ).
6. Repeat steps 2 to 5 until  $|Y_{j,t}| = Y^+$  or until the matrix is empty.

Where  $Y^+ > 0$  is a cap on the number of car variants a manufacturer can simultaneously offer. This cap suggests that manufacturers are constraint in their degree of diversification due to strategic considerations such as brand-reputation.

### A.2.3 Innovation

With probability  $\iota \in (0, 1)$ , manufacturers seek innovations to expand the set of known technologies, reflecting that innovation attempts may not be successful in becoming a novel technology (Nelson and Winter, 1977; Dosi et al., 2010). When innovating, manufacturers build a list of candidate technologies to develop ( $M_{j,t}^{RD}$ ). The list of candidate technologies is formed by the unexplored technologies at a local distance (1 Hamming distance from the NK coordinates) with respect to the last explored technology in the ICEV ( $m_{j,t}^{ICE} \in M^{ICE}$ ) and the EV ( $m_{j,t}^{EV} \in M^{EV}$ ) landscapes. Additionally, the set of candidate technologies includes the current search locations on each landscape to allow firms to not research if the alternative candidate technologies render insufficient profit. Equation ?? formally defines the list of candidate technologies:

$$\begin{aligned} M_{j,t}^{RD} = & \left\{ m \in M^{EV} \setminus M_{j,t-1} : \sum_{n=0}^N |\phi_{n,m'} - \phi_{n,m}| = 1 \right\} \\ & \cup \left\{ m \in M^{ICE} \setminus M_{j,t-1} : \sum_{n=0}^N |\phi_{n,m''} - \phi_{n,m}| = 1 \right\} \\ & \cup \{m', m''\} \end{aligned} \quad (10)$$

Where  $\phi_{n,m}$  defines the coordinates of a technology  $m$  on each of the  $n$  dimensions of the  $NK$  landscape.

The probability of selecting a technology from the list of candidate technologies is proportional to its relative expected profit. The variable  $\Pi_{m,t}^*$  returns the expected profit assigned to a technology  $m$  optimally priced in its most profitable segment. Denoting the selected researched technology with  $m_{j,t}^{RD}$ , Equation ?? defines its selection probability, conditional on probability  $\iota$ :

$$Pr(m_{j,t}^{RD} = m' \in M_{j,t}^{RD}) = \frac{\exp(\lambda \Pi_{m',j,t}^*)}{\sum_{m=0}^{M_{j,t}^{RD}} \exp(\lambda \Pi_{m,j,t}^*)} \quad (11)$$

Where the parameter  $\lambda > 0$  indicates the research intensity of choice. The lower the  $\lambda$ , the less likely a firm is to get stuck in a local solution, but also the less likely a firm selects the best solution. This method has been widely used in the literature as a way to balance exploration and exploitation, which is a common issue in complex search problems (Puranam et al., 2015).

Following this step, the search locations in each landscape are updated to the newest car technology researched,

$$\begin{aligned} m_{j,t}^{EV} &= \begin{cases} m_{j,t}^{RD} & \text{if } m_{j,t}^{RD} \in M^{EV} \\ m_{j,t-1}^{EV} & \text{if } m_{j,t}^{RD} \notin M^{EV} \end{cases} \\ m_{j,t}^{ICE} &= \begin{cases} m_{j,t}^{RD} & \text{if } m_{j,t}^{RD} \in M^{ICE} \\ m_{j,t-1}^{ICE} & \text{if } m_{j,t}^{RD} \notin M^{ICE} \end{cases} \end{aligned} \quad (12)$$

Finally, the searched technology is added to the set of known technologies. If the firm exceeds its memory capacity ( $M^+ > 0$ ), the oldest technology to be searched not in production is forgotten. Firm's limited memory capacity is a cognitive limitation whereby the searching agent economizes cognitive power by forgetting unused solutions. This limitation has been widely studied in heuristic search algorithms inspired in animal behaviour (Tang et al., 2012). We define a timer  $T_{m,j,t}$  in Equation ?? for this purpose:

$$T_{m,j,t}(m \in M_{j,t}) = \begin{cases} T_{m,j,t-1} + 1 & \text{if } m \notin Y_{j,t} \\ 0 & \text{if } m = m_{j,t}^{RD} \vee y \in Y_{j,t} \end{cases} \quad (13)$$

Hence, the set of known technologies is expressed as in Equation ??:

$$M_{j,t} = \begin{cases} M_{j,t-1} & \text{if } m_{j,t}^{RD} \in M_{j,t-1} \\ M_{j,t-1} \cup \{m_{j,t}^{RD}\} \setminus \{\operatorname{argmax}_{m \in M_{j,t-1}} T_{m,j,t}\} & \text{if } |M_{j,t-1}| = M^+ \\ M_{j,t-1} \cup \{m_{j,t}^{RD}\} & \text{if } |M_{j,t-1}| < M^+ \end{cases} \quad (14)$$

#### A.2.4 Used cars

The market for used cars represents the bulk of car purchases in the US<sup>2</sup>, providing an affordable alternative to low-income users.

This model chooses a parsimonious approach for the representation of the market for used cars. Used car dealers are consolidated into a single entity that prices cars based on the market price of the most similar car in the market for new cars, thus implicitly accounting for market segmentation.

When considering buying a car  $v \in V_t : v_{v,t} \neq 0$ , the second-hand dealer targets a markup  $\mu > 0$ . Despite the simplification that used car dealers are consolidated into a single entity, the parameter  $\mu$  serves as a proxy for degree of monopsony in the market. Thus, under perfect competition ( $\mu = 0$ ), the used car dealer buys a car at a price equal to its estimated selling price, denoted by  $\mathbb{E}(P_{i,t})$ , whereas a perfect monopsony ( $\mu = \infty$ ) buys cars at their scrapping value, which we assume to be a fixed constant  $F > 0$ . If the estimated sale price of a car is lower than its scrapping value ( $F$ ), the buying price becomes the scrapping value. Equation ?? defines the price the second hand merchant offers for a car  $v$ :

$$\hat{P}_{v,t} = \operatorname{MAX} \left( F, \frac{\mathbb{E}(P_{i,t})}{(1 + \mu)} \right) \quad (15)$$

For the calculation of the expected market price, the used car dealer uses as a reference the price of the car in the new market with the most similar features. The reference price is denoted by  $P_{v,t}^T$  and defined as in Equation ??:

$$P_{v,t}^T = P_{y',t} \quad (16)$$

$$y' = \operatorname{argmin}_{y \in Y_{t-1}} \left| \frac{\omega_{v,t} - \omega_{y,t-1}}{\operatorname{MAX}(\omega_{y,t-1} : y \in Y_{t-1})} \right| + \left| \frac{Q_{v,t} - Q_{y,t-1}}{\operatorname{MAX}(Q_{y,t-1} : y \in Y_{t-1})} \right| + \left| \frac{B_{v,t} - B_{y,t-1}}{\operatorname{MAX}(B_{y,t-1} : y \in Y_{t-1})} \right| \quad (17)$$

Where car features are normalized to ensure both dimensions are in the same order of magnitude. The expected market price of a car owned by an individual  $i$  is a fraction of its reference price. This fraction is given by its monthly type-specific price depreciation ( $\xi_{v,t}$ ) (Schloter, 2022), as Equation ??:

$$\mathbb{E}(P_{v,t}) (v_{v,t} \neq 0) = P_{i,t}^T (1 - \xi_{v,t})^{L_{v,t}} \quad (18)$$

The quality and price depreciation monthly rates feature different values because price adjustments do not map one-to-one the quality decay. Given our parsimonious representation of the second-hand market, we parametrize with  $\mu$  and  $\xi$  the off-model dynamics.

Likewise, Equation ?? defines the actual sale price of a used car:

$$P_{v,t} (v_{v,t} = 0) = P_{i,t}^T (1 - \xi_{v,t})^{L_{v,t}} \quad (19)$$

The used car dealer scraps all cars whose sale price is lower than the scrapping value. Therefore, the set of active cars  $V_t$  increases with the purchases of new cars and decreases with the scrapping

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<sup>2</sup>Statista

of old cars, as defined in Equation ??<sup>3</sup>:

$$V_t = V_{t-1} \bigcup_{i=1}^I \{\psi_{i,t} \in Y_t\} \setminus \{v \in V_{t-1} : P_{v,t} < F, v_{v,t} = 0\} \quad (20)$$

## B Model Parameters and Variables

Table 2: Parameters

| Symbol   | Name                                       | Description                                                                                                              | Value          | Units       | Source                             |
|----------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------|-------------|------------------------------------|
| $\eta$   | Car purchase consideration probability     | Probability that a car user considers purchasing a car.                                                                  | 8.3            | Percent     | -                                  |
| $r$      | Discount rate                              | Monthly discount rate used to derive the consumer surplus.                                                               | 0.165          | Percent     | (Greene, 2010; Busse et al., 2013) |
| $\alpha$ | Quality diminishing returns factor         | Factor controlling the diminishing returns to scale for the contribution of car quality to the total willingness to pay. | 0.5            | -           | -                                  |
| $\zeta$  | Range diminishing returns factor           | Factor controlling the diminishing returns to scale for the contribution of car range to the total willingness to pay.   | 0.296          | -           | (Ferguson et al., 2018)            |
| $\beta$  | Willingness to pay for quality             | Monetary value an individual assigns to car quality.                                                                     | Median 208,278 | USD/Quality | Indirect calibration               |
| $\nu$    | Willingness to pay for range               | Monetary value an individual assigns to car range.                                                                       | 1173.65        | USD/km      | (Ferguson et al., 2018)            |
| $\gamma$ | Willingness to pay for emissions reduction | Monetary value an individual assigns to CO2 emission reductions.                                                         | Median 0.066   | USD/kgCO2   | (Hulshof and Mulder, 2020)         |
| $D_i$    | Distance driven                            | User-specific monthly distance driven.                                                                                   | Mean 1119.84   | km/month    | California Energy Commission       |

<sup>3</sup>To reduce computational cost we impose a maximum capacity of  $V^+$  the second hand market, nonetheless the market size remains an order of magnitude larger than the number of second hand cars sold per time step ensuring ample selection choices, and we scrap all cars that have not been sold after  $T^+$  periods

Table 2: Parameters

| Symbol    | Name                          | Description                                                               | Value                              | Units        | Source                            |
|-----------|-------------------------------|---------------------------------------------------------------------------|------------------------------------|--------------|-----------------------------------|
| $\chi_i$  | EV Im-itation threshold       | Threshold representing an individual's peer influence in considering EVs. | Beta-distribution (1.1968, 2.6805) | Percent      | Indirectly calibrated             |
| $\kappa$  | Consumer intensity of choice  | Parameter indicating the intensity of choice in the logit model.          | 2.25e-4                            | -            | Indirectly calibrated             |
| $\iota$   | Innovation probability        | Probability that a manufacturing firm attempts to innovate.               | 8.3                                | Percent      | -                                 |
| $\theta$  | Production change probability | Probability that a firm updates its supply line and prices.               | 8.3                                | Percent      | -                                 |
| $\lambda$ | Research intensity of choice  | Parameter indicating the intensity of choice in the innovation model.     | 1e-3                               | -            |                                   |
| $M^+$     | Technology memory capacity    | The maximum number of known technologies a firm can remember.             | 30                                 | Technologies | -                                 |
| $Y^+$     | Car variant limit             | The maximum number of car types a firm can produce.                       | 10                                 | Technologies | -                                 |
| $T^+$     | Used car time limit           | Maximum number of months before a car gets scrapped.                      | 36                                 | Months       | -                                 |
| $V^+$     | Used car stock limit          | Maximum number used cars in the used car market.                          | 300                                | Cars         | -                                 |
| $\mu$     | Used car dealer markup        | Target markup applied by used car dealers when pricing used cars.         | 100                                | Percent      | -                                 |
| $F$       | Car scrapping value           | Scrapping value of a car.                                                 | 669.80                             | USD          | <a href="#">JunkCarsUS</a>        |
| $\xi$     | Price depreciation            | Monthly type-specific depreciation rate for the price of used cars.       | ICEV: 1.16, EV: 0.87               | Percent      | (Schloter, <a href="#">2022</a> ) |
| $\delta$  | Fuel efficiency depreciation  | Monthly depreciation rate for the car fuel efficiency.                    | 0.175                              | Percent      | Indirect calibration              |

Table 2: Parameters

| Symbol                 | Name                                 | Description                                                                                                     | Value                                | Units                    | Source                                  |
|------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------|-----------------------------------------|
| $N$                    | Number of dimensions in NK landscape | Number of dimensions (components) in the NK landscape representing technologies.                                | 15                                   | Number of components     | -                                       |
| $K$                    | Epistasis                            | Epistasis in the NK landscape, representing the degree of interdependency between components of the technology. | 3                                    | Number of components     | -                                       |
| $E$                    | Manufacturing emissions              | Type-specific manufacturing emissions.                                                                          | ICEV: 10<br>EV: 14                   | Tonnes of CO2            | (Jaramillo et al., 2022, p. 1076)       |
| $(C^-, C^+)$           | Manufacturing costs                  | Maximum and minimum manufacturing costs in the NK landscape.                                                    | (5,7)                                | Thousand USD             | (Grieco et al., 2024)                   |
| $(Q^-, Q^+)$           | Car quality                          | Maximum and minimum quality in the NK landscape.                                                                | (0,1)                                | Abstract quality measure | -                                       |
| $\rho(C, Q)$           | Correlation cost-quality             | Correlation coefficient between manufacturing costs and car quality in the NK landscape.                        | (0)                                  | Correlation coefficient  | -                                       |
| $(\omega^-, \omega^+)$ | Fuel efficiency                      | Maximum and minimum type-specific fuel efficiency in the NK landscape.                                          | ICEV: (0.79,3.09)<br>EV: (2.73,9.73) | km/KWh                   | (Jaramillo et al., 2022, Appendix 10.2) |
| $\rho(C, \omega)$      | Correlation cost-efficiency          | Correlation coefficient between manufacturing costs and fuel efficiency in the NK landscape.                    | (0)                                  | Correlation coefficient  | -                                       |
| $(B^-, B^+)$           | Fuel tank / Battery size             | Maximum and minimum fuel tank (ICEV) or battery size (EV) in the NK landscape.                                  | ICEV: (469.4)<br>EV: (0,150)         | KWh                      | (Schmuck et al., 2018)                  |
| $\rho(C, B)$           | Correlation cost-quality             | Correlation coefficient between manufacturing costs and fuel tank or battery size in the NK landscape.          | ICEV: 0,<br>EV: .5                   | Correlation coefficient  | -                                       |



Table 2: Parameters

| Symbol      | Name                                 | Description                                                                                   | Value                           | Units                    | Source                                                                                   |
|-------------|--------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------|--------------------------|------------------------------------------------------------------------------------------|
| $C_m$       | Manufacturing costs                  | Unit production costs of technology $m$ at time $t$ .                                         | Technology-specific             | USD                      | NK model                                                                                 |
| $Q_m$       | Quality                              | Catch-all quality measure of car using technology $m$ .                                       | Technology-specific             | Abstract quality measure | NK model                                                                                 |
| $B_m$       | Battery/fuel tank size               | Size of the battery or fuel tank of car using technology $m$ .                                | Technology-specific             | kWh                      | NK model                                                                                 |
| $\omega_m$  | Fuel efficiency                      | Fuel or energy efficiency of car using technology $m$ .                                       | Technology-specific             | km/kWh                   | NK model                                                                                 |
| $c_{v,t}$   | Fuel cost                            | Cost per kilowatt hour of gasoline or electricity                                             | Time-variant                    | USD/KWh                  | ICEV: <a href="#">EIA</a><br>EV: <a href="#">EIA</a>                                     |
| $e_{v,t}$   | Fuel emissions                       | CO2 emissions per kilowatt hour of gasoline or electricity                                    | ICEV: 0.331<br>EV: Time-variant | kgCO2/KWh                | ICEV: (Jaramillo et al., <a href="#">2022</a> , Table 10.8)<br>EV: <a href="#">EMBER</a> |
| $J$         | Number of firms                      | Number of firms in the simulation.                                                            | 16                              | Number of firms          | (Grieco et al., <a href="#">2024</a> )                                                   |
| $I$         | Number of car users                  | Number of car users in the simulation.                                                        | 3000                            | Number of car users      | (Grieco et al., <a href="#">2024</a> )                                                   |
| $R^\beta$   | Number of quality-based segments     | Number of partitions used to create market segments based on the distribution of $\beta_i$ .  | 8                               | -                        | -                                                                                        |
| $R^\gamma$  | Number of environment-based segments | Number of partitions used to create market segments based on the distribution of $\gamma_i$ . | 2                               | -                        | -                                                                                        |
| $R^\nu$     | Number of range-based segments       | Number of partitions used to create market segments based on the distribution of $\nu_i$ .    | 1                               | -                        | -                                                                                        |
| $\rho_{ws}$ | Network density                      | Density connections between car users in Watts-Strogatz network                               | 0.05                            | -                        | -                                                                                        |

Table 2: Parameters

| Symbol   | Name                  | Description                                                                              | Value | Units | Source |
|----------|-----------------------|------------------------------------------------------------------------------------------|-------|-------|--------|
| $p_{WS}$ | Re-wiring probability | Probability of re-wiring close range links to long range links in Watts-Strogatz network | 0.1   | -     | -      |

Table 3: Variables

| Symbol           | Name                                  | Description                                                                                            | Units            |
|------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------|------------------|
| $Y_t$            | Set of new cars on sale               | Set of new cars on sale by all firms at time $t$                                                       | -                |
| $Y_{j,t}$        | Firm-specific set of new cars on sale | Set of new cars on sale by firm $j$ at time $t$                                                        | -                |
| $V_t$            | Set of active cars                    | Set of all active cars owned by users or in the used car market at time $t$                            | -                |
| $A_{i,t}$        | Set of considered alternatives        | Set of car alternatives considered by individual $i$ at time $t$ .                                     | -                |
| $z_{v,t}$        | Car type                              | Binary statement specifying if a car $v$ is an electric vehicle.                                       | Binary statement |
| $v_{v,t}$        | Car Owner ID                          | Identifier of the owner of car $v$ at time $t$ (0 if in the second-hand market).                       | User ID          |
| $U_{a,i,t}$      | Consumer utility                      | Consumer surplus for car alternative $a$ for individual $i$ at time $t$ .                              | USD              |
| $\delta$         | Fuel efficiency depreciation rate     | Fuel or energy efficiency monthly depreciation rate.                                                   | Percent          |
| $L_{a,t}$        | Car age                               | Age of car $a$ at time $t$ .                                                                           | Months           |
| $LCE_{a,i,t}$    | Lifecycle emissions                   | Lifecycle emissions of car $a$ for individual $i$ at time $t$ .                                        | kgCO2            |
| $LCC_{a,i,t}$    | Lifecycle cost                        | Lifecycle cost of car $a$ for individual $i$ at time $t$ .                                             | USD              |
| $P_{a,t}$        | Purchase cost                         | Purchase cost of new or used car $a$ at time $t$ .                                                     | USD              |
| $\hat{P}_{i,t}$  | User car sale price                   | Sale price of user $i$ 's car on the second-hand market at time $t$                                    | USD              |
| $\psi_{i,t}$     | Chosen car                            | ID of individual $i$ 's chosen car at time $t$ .                                                       | Car ID           |
| $H_i$            | Individual $i$ 's network             | Set of car users representing $i$ 's network neighbourhood.                                            | -                |
| $V_{i,t}^H$      | Network neighbours cars               | Subset of active car that belong to car users in $i$ 's network                                        | -                |
| $x_{i,t}$        | EV consideration                      | Binary variable indicating whether individual $i$ considers buying EVs at time $t$ .                   | Binary statement |
| $M_{j,t}$        | Set of known technologies             | Set of technologies known to firm $j$ at time $t$ .                                                    | -                |
| $\Pi(P)_{m,s,t}$ | Estimated segment-specific profit     | Estimated profit a firm expects to make in segment $s$ using technology $m$ at price $P$ at time $t$ . | USD              |

Table 3: Variables

| Symbol Name     | Description                                                                                                                                                                                           | Units  |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| $W_{s,t}$       | Expected segment-specific state of competition<br>Expected aggregate consumer surplus in sector $s$ at time $t$ .                                                                                     | USD    |
| $P_{s,m,t}^*$   | Optimal price<br>Optimal price of a technology $m$ in segment $s$ at time $t$                                                                                                                         | USD    |
| $M_{j,t}^{RD}$  | Set of candidate technologies to explore<br>Set containing all the unexplored neighbouring technologies with respect to the past search locations on each landscape of firm $j$ at time of firm $t$ . | -      |
| $m_{j,t}^{RD}$  | Searched technology<br>Searched technology by firm $\mathbf{j}$ at time $\mathbf{t}$ .                                                                                                                | -      |
| $m_{j,t}^{EV}$  | Searched location in EV landscape<br>Searched location in EV landscape by firm $\mathbf{j}$ at time $\mathbf{t}$                                                                                      | -      |
| $m_{j,t}^{ICE}$ | Searched location in ICE landscape<br>Searched location in ICE landscape by firm $\mathbf{j}$ at time $\mathbf{t}$                                                                                    | -      |
| $\vec{\phi}_m$  | Technology coordinates<br>Coordinates of technology $m$ in the NK landscape.                                                                                                                          | -      |
| $T_{m,j,t}$     | Memory timer<br>Number of periods at time $t$ since the last time a technology $m$ has been explored or used by firm $j$ .                                                                            | Months |
| $P_{v,t}^T$     | Reference price<br>Reference price in the market for new cars to be used as a benchmark when pricing used car $v$ at time $t$                                                                         | USD    |
| $I_{s,t}$       | Consumer segment set<br>Set with users belonging to market segment $s$ at time $t$ .                                                                                                                  | -      |

## C Additional Figures

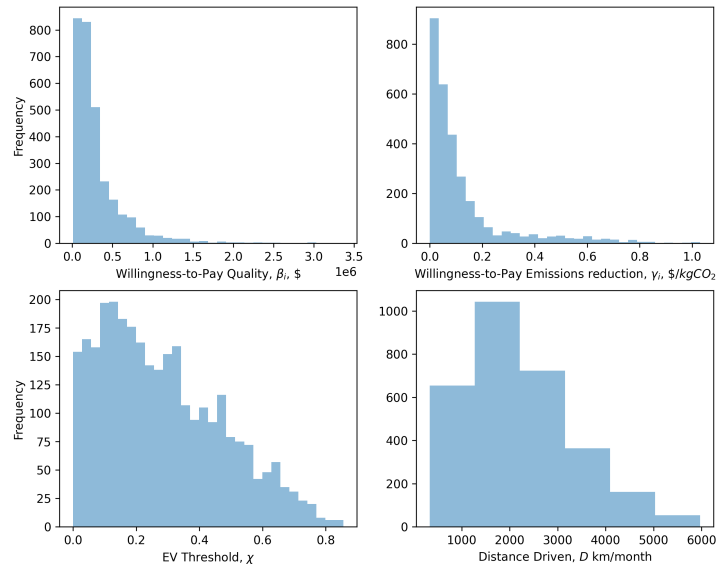


Figure 3: Histogram of willingness to pay for quality (top left), willingness to pay for emissions reduction (top right), EV imitation threshold (bottom left), distance driven (bottom right).